Fiscal Year:	FY 2011	**Task Last Updated:**	FY 08/15/2011
PI Name:	Blakely, Eleanor A Ph.D.		
Project Title:	Early Markers of Space-Radiation Induced Human Cataractogenesis		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
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Flight Program:			
Flight Assignment:	NOTE: New end date is 12/22/2010 per C. Guidry/JSC (10/2010) NOTE: New end date is 9/30/2010 per C. Guidry/JSC (6/2010)		
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COI Name (Institution):	Chang, Polly (SRI International)		
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Task Description:	The proposed project is a competitive renewal application with research emphasis focused on reducing the risk of radiation-induced cataract in human space travel. A clear understanding of the underlying mechanisms for cataractogenesis is necessary for early diagnosis and mitigation of cataract risks. The hypothesis driving our research is that particle radiation induces early molecular signaling alterations in the lens epithelial cells, which disrupt normal differentiation mechanisms necessary for the maintenance of lens clarity. In order to assess the time course of the expression and the natural history of the developing lesion, two experimental models of the lens are proposed in this application. Differentiating human lens cells in vitro will be used to extend the investigation of molecular mechanisms of action that follow particle radiation induced changes in key marker proteins using beams defined by NASA's operational parameters, e.g. low doses and dose-rates of protons or heavy ions. Mice will be irradiated and early lenticular changes associated with molecular aggregation will be followed using an exquisitely sensitive dynamic light scattering method to map the unique signatures of particle radiation-induced lens opacification. Information gathered from these measurements will be used to determine when the lenses will be harvested for protein identification. The results of the proposed research will improve mechanistic understanding underlying variability of dose-rate dependencies for cataract formation, and reduce the uncertainties in cataract risk assessment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of our work will be used to pursue countermeasures that could reduce the risk of cataract. These studies may also contribute to estimations of risk assessment for radiation-induced cataracts among astronauts exposed to particles during space travel, especially during missions into deep space. We may also contribute to the development of countermeasures for radiation-induced cataracts from other occupational and clinical radiation exposures. The biological mechanisms underlying normal lens fiber cell differentiation may also be elucidated. The exact factor(s) responsible for the continued replication of lens epithelial cells in embryos and adults remains to be determined. These studies therefore should shed new light on the process of human lens cell differentiation that has been difficult to explore with animal models in vivo or with human lens cultures in vitro that are either immortalized and/or undergo only limited differentiation.
Task Progress:	The long-term objectives of our project are to elucidate the underlying molecular basis for heavy-ion-induced radiation cataractogenesis, and to obtain a more complete understanding of the normal process of lens cell differentiation. The weight of experimental evidence indicates that the germinative epithelium is the primary target for radiation damage leading to cataract. However, there are potentially three hypotheses for the mechanism of radiation-induced cataractogenesis: 1) the increased genotoxic load of radiation damage leads to cataract through a number of intermediate steps involving altered expression, 2) gene expression is altered without genomic changes (the effect here may be at the level of signaling), or 3) the effect is on protein expression directly. There is of course the possibility that these three hypotheses are not mutually exclusive, and that some combination of these hypotheses is involved. We postulate that cytokines play a role in driving the cell-cycle regulation in lens epithelial cells. The cascades of signals trigger multiple membrane-based events, including alterations in actin cytoskeletal dynamics, integrin extracellular matrix expression, adhesion complex remodeling, intra- and inter-cellular communication, and cell movement. For homeostasis, these interactions are tightly coupled. Using a high throughput quantitative approach to query selected genes known to be important for extracellular communication, we show dynamic changes in gene expression profile including characteristic differences in both the number and distribution of genes in our in vitro differentiating human lens epithelial cells system (HLE) after a low- or high-dose of either low- or high-LET, in a short time course up to 8 h after exposure. We noted that the profile of expression of Extracellular Matrix (ECM) genes was dependent not only on the dose of radiation and sampling times, but also on the radiation quality of the beams. The expression of more than 20 ECM genes, dominated by ECM proteases, cell matrix adhesion and transmembrane molecules, were upregulated within 4 hr after exposure to low doses of protons. However, these changes were transient and did not persist for more than 8 hr post irradiation. On the other hand, more ECM genes were impacted after 100 cGy of protons and the expression persisted beyond 8 hrs post irradiation. In sharp contrast with the ECM expression profile after protons, fewer genes had altered gene expression patterns after exposure to titanium ions when compared to the controls. In particular, genes that were affected were primarily down regulated in the 1, 4 and 8 h time points studied. We examined the expression profile of a similar set of extraceullar matrix and adhesion molecules in whole lenses from Sprague-Dawley rats collected nine months after exposure to 10 or 100 cGy of 600 MeV/u iron ions. The results were compared to a set of control sham-treated lenses that were confirmed to be non-cataractous. Detailed visual inspection of iron-ion irradiated lenses at harvest confirms that the cataractous lesions are multifocal and polymorphic, with anterior cortical opacification, and anterior cortical involvements. Opacities appear to aggregate in the equatorial periphery. In addition, posterior subcapsular cataractous (PSC) dots are clearly visible in 50% of the lenses that were examined ex vivo. Results from focused microarray analysis revealed that lenses from 10 cGy-iron-ion-irradiated rat lenses showed significant 3- to 4-fold increases in genes in CAM & ECM functional groups in lenses. On the other hand, 100 cGy-iron-ion-irradiated rat lenses show a completely different gene response. Only 2 genes in the ECM & CAM series were increased 2- to 3-fold in the high-dose irradiated cataractous lenses, and two CAM series genes were significantly down-regulated 2- to 3-fold nine months after exposure. The high responsiveness of ECM gene families after a single low radiation dose to the lens in vivo contrasts with the decreased gene expression seen after a 10-fold higher radiation dose. The fact that such changes were persistent nine months after the exposures is unexpected. Clearly, different genes are affected at each dose level. It is not yet known how this gene profile correlates with the status of the opacifications observed at each dose at the same time post-exposure. CD44, a cell-surface glycoprotein and a member of the cell adhesion molecule was shown to be a candidate that was upregulated at 8h after 100 cGy protons. Misregulation of this gene was also found in rat lenses >9 months after 10 cGy iron ions. CD44 in rat lenses have previously been reported and surgical removal of lens fiber cells from adult mice often results in a robust upregulation of CD44 protein, which preceded the upregulation of alpha-smooth muscle actin expression typically used as a marker of epithelial-mesenchyma transition in this model of lens epithelial cell fibrosis. CD44 protein has also been shown to increase in the lens epithelium after injury and is rodent-strain dependent. The CD44 family of receptors includes multiple variant isoforms due to post-translational modification and alternative splicing. Several isoforms have been linked to malignant properties, including migration, invasion and metastasis. Our Western results with isoform specific antibodies to probe for translation of CD44 variant gene product suggest that although CD44 protein levels were reduced 1 h after a small 10 cGy of protons, the v6 variant levels remained similar to the control levels. On the other hand, while the CD44 levels were elevated at 1.5 h after 100 cGy Titanium, the v6

	variant also remained similar to the control levels. C57Bl6 male mice were sham-treated or exposed to 10, 50, 100 and 200 cGy of 600 MeV/u iron or 1 GeV/u protons in two separate large scale studies during NSRL08C and 09C. The animals were maintained for a period of 12 – 18 months post irradiation to measure (1) body weight changes, (2) changes in lens clarity using (a) slit lamp, and (b) quasi-elastic light scattering (QLS) technique. Lens tissues were also ex vivo imaged using stereo microscopy to document lens pathology and preserved for molecular analysis. Our collaborator Dr. Lee E. Goldstein has developed an analytical method to deconvolute the non-invasive infrared quasi-elastic light scattering and intensity fluctuation measurements to generate an autocorrelation function tau. The time constant for tau is an indicator of the size of the protein aggregation and light scatting centers in the lens. To date, comparisons have been made to show if the early in vitro human lens cell responses within hours after exposure correlate with the persistent in vivo whole rat lens response profile nine months after low doses. A comprehensive investigation of the dose-dependent expression of ECM and cell adhesion molecules (CAM) genes in rat, murine and human lens cells is ongoing for final preparation of a manuscript comparing the responses in three species.
Bibliography Type:	Description: (Last Updated: 02/12/2026)
Articles in Peer-reviewed Journals	Blakely EA, Kleiman NJ, Neriishi K, Chodick G, Chylack LT, Cucinotta FA, Minamoto A, Nakashima E, Kumagami T, Kitaoka T, Kanamoto T, Kiuchi Y, Chang P, Fujii N, Shore RE. "Radiation cataractogenesis: epidemiology and biology." Radiat Res. 2010 May;173(5):709-17. PubMed PMID: 20426671 , May-2010